

The Interpretive Graph Overlay: Modeling Case Law as Normative Events for Deterministic Legal Retrieval

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Abstract

Legal retrieval systems reliably recover the literal wording of statutes but often miss the *applicable interpretation* when courts revise their reading over time. This demonstration presents the *Interpretive Graph Overlay*, an extension of SAT-Graph that models judicial decisions as Action entities generating semantically distinct **Interpretive Versions** of norms, encoding their *ratio decidendi*, scope, and validity interval. By reusing SAT-Graph’s primitives (Item, Version, Action, TextUnit), the overlay enables deterministic, time-aware retrieval: agents answer not only “What does the law say?” but also “How was it interpreted on this date?” We demonstrate the approach through JSON encodings of three landmark Brazilian Supreme Court cases.

CCS Concepts

• Applied computing → Law; • Computing methodologies → Knowledge representation and reasoning.

Keywords

Legal Knowledge Graphs, GraphRAG, Case Law Modeling, Neuro-Symbolic AI, Judicial Interpretation

1 Introduction and Motivation

Legal norms are targets in motion, subject to *interpretive drift*: a statute’s text may remain immutable while the applicable norm is radically altered. The U.S. Due Process Clause has been textually static since 1868, yet the prevailing norm on abortion was inverted between *Roe* and *Dobbs*. In civil law jurisdictions, courts recognize “constitutional mutations” — changes to binding interpretation without legislative amendment. Standard RAG architectures, and even frameworks like SAT-Graph [7] that reconstruct statutory wording at any historical date, are blind to this interpretive layer.

This demonstration introduces the *Interpretive Graph Overlay*, which models judicial decisions not as passive documents but as Action entities that generate new **Interpretive Versions** of the norms they construe. The overlay extends SAT-Graph by generalizing its abstractions: just as a legislative amendment creates a new textual version of an article, a binding judgment creates an interpretive version that inherits the statute’s text but carries a distinct *ratio decidendi* and a defined validity interval. The result is a unified knowledge graph where a query for a provision returns both the legislative text and the prevailing judicial interpretation.

Related Work. The temporal dimension of legal concepts has a long tradition in AI & Law. Rissland and Friedman [14] proposed detecting change in legal concepts through case sequences. Hafner and

Berman [9] argued that teleological, temporal, and procedural context is essential to case-based legal reasoning. More recently, Henderson and Bench-Capon [5, 10] modeled the development of case law as a dialogical process in which successive decisions refine and contest earlier rationales. Complementary work by Araszkievicz, Francesconi, and Zurek [3] integrates semantic-web statutory modeling with the representation of multiple coexisting interpretations. Legal ontologies such as LKIF-Core [11] and Akoma Ntoso [12], and case-based systems from HYPO [4] and CATO [1] to Araszkievicz’s *Case Frames* [2], have largely kept statutes and precedents in separate representational silos. GraphRAG approaches [8] improve retrieval over graphs but lack a model of legal validity.

Positioning. The overlay is not a dynamics-of-case-law engine in the dialogical sense of Henderson and Bench-Capon; it is a *persistence and retrieval substrate* on which such engines can be layered. Where Rissland and Hafner emphasize the detection and contextualization of change, the overlay assumes that change has been authoritatively established and focuses on making its temporal footprint deterministically queryable. Coexisting interpretations, in the spirit of Araszkievicz, Francesconi, and Zurek, are represented natively as parallel Interpretive Versions over the same statutory item.

2 Ontological Parsimony

Rather than introducing ad-hoc entities for judicial reasoning, the overlay leverages SAT-Graph’s event-centric architecture, treating legislative and judicial acts as structurally equivalent: both produce temporal versions through discrete actions. A **Temporal Version (TV)** represents the state of a norm after an event; when the event is a binding precedent, the result is an **Interpretive Temporal Version (I-TV)**. The mechanism is identical in both cases (Figure 1): the legal document (statute or judgment) is a versioned *Instrument*; the specific **micro-event** (legislative or judicial command) is an Action that uses the instrument, modifies the target’s previous version, and creates a new derived version.

Statutory Versions (S_TV) and Interpretive Versions (I_TV) co-exist as ontological peers in the same derivation chain, allowing a single traversal engine to interleave legislative amendments with judicial interpretations. To isolate the binding *ratio decidendi* from *obiter dicta*, judgments are decomposed into citable components (opinions, votes, rulings) following SAT-Graph’s three-tier pattern; the Action may link fine-grained to a specific component or coarse-grained to the judgment as a whole.

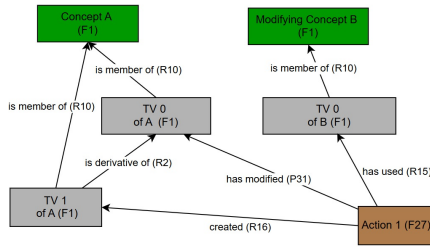


Figure 1: The isomorphic event topology. The Action is the causal nexus linking instrument, previous version, and newly derived version. The topology is invariant across statutory and interpretive events.

3 Semantic Mapping and Artifact

The overlay relies on four SAT-Graph primitives [6]: Item (abstract identity of a legal artifact), Version (its temporal state), Action (the causal micro-event), and TextUnit (a concrete textual realization). A single Version can carry multiple TextUnits, each tagged with a semantic aspect (facts, ratio, holding). This decouples logical argumentation from document structure: the binding reasoning of a Brazilian Supreme Court judgment, for instance, maps to the *Voto* component as a TextUnit with aspect=ratio, regardless of its location in the document hierarchy. Listing 1 shows the resulting serialization for a mutation of the presumption of innocence.

```

{
  "id": "urn:lex:br:federal:constituicao:...@2016-02-17!art5_inc57_hc126292",
  "item_id": "urn:lex:br:federal:constituicao:1988-10-05;1988@art5_inc57",
  "type": "interpretive",
  "validity_interval": { "start": "2016-02-17", "end": "2019-11-06" },
  "metadata": { "version_type": "interpretive_mutation" },
  "text_units": [
    { "aspect": "ratio", "language": "en", "content": "The provisional..." },
    { "aspect": "holding", "language": "en", "content": "Imprisonment is ..." }
  ]
}

```

Listing 1: Interpretive Version generated by HC 126.292.

The primitive `getValidVersions(item_id, date)` operates polymorphically, returning three possible patterns: *stability* (a single S_{TV} applies), *interpretive overlay* (the S_{TV} is modified by an I_{TV}), and *interpretive divergence* (the S_{TV} coexists with conflicting I_{TV} s, representing jurisprudential splits).

4 Demonstration

The demonstration consists of three artifacts presented together at the poster: (i) structured JSON encodings of each interpretive version, following the serialization of Listing 1 and available in the companion repository; (ii) printed graph topologies materializing the Action, Version, and TextUnit relationships for each case; and (iii) walkthrough slides tracing the execution of `getValidVersions` over the encodings for concrete query dates. These artifacts show how the overlay represents interpretive drift end-to-end, from the judicial event to the retrievable *ratio decidendi*.

The running example is the Brazilian constitutional text on the presumption of innocence (Art. 5, LVII), textually unchanged since 1988 yet subject to two binding interpretive mutations. For a query dated 2017-06-01 (“Can a defendant be imprisoned after conviction in second instance?”), `getValidVersions` returns both S_{TV}_{1988}

and I_{TV}_{2016} (from HC 126.292); the agent retrieves the ratio aspect of I_{TV}_{2016} and answers “yes,” citing the precedent despite the unchanged text. A query dated 2020-06-01 returns instead I_{TV}_{2019} (from ADCs 43, 44, 54), flipping the answer without any textual amendment. Two further case studies in the repository illustrate *constitutional dialogue* (ADI 4983 on *Vaquejada*, followed by Constitutional Amendment 96/2017) and *simultaneous divergence* (conflicting IPI credit rulings from two Supreme Court Panels).

5 Conclusion

The Interpretive Graph Overlay extends SAT-Graph to case law without introducing domain-specific entities, reusing the event-centric primitives of statutory versioning. This parsimony yields explainable, time-aware retrieval with traceable citations. The current design prioritizes precision and auditability: *implicit* overruling is left to the ingestion layer rather than encoded as graph-level inference. Ongoing work targets quantitative evaluation against LegalBench-RAG [13] and LLM-assisted ingestion pipelines for larger jurisdictional corpora.

Data and Artifact Availability

The structured JSON encodings of the three case studies (HC 126.292, ADI 4983, and the IPI divergence) are available at: <https://github.com/hmartim/sat-graph-api>.

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